Lab 2 – Smart City Requirements Specifications (SRS)

Joshua Connelly

Old Dominion University

CS411W

Sumaya Sanober

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1. Introduction

1.1. Purpose

This document defines the functional and system requirements for the **Smart City** software system. Its primary audience is the development team responsible for implementing, testing, and maintaining the system. This SRS provides a detailed technical reference that specifies system features, architecture, interfaces, user roles, and dependencies. It ensures a common understanding of the system requirements before development begins.

1.2. Scope

The **Smart City** system is a mobile and web-based application that provides **real-time urban navigation**, **location-based community forums**, and **user-generated insights** to improve city mobility.

Included Features:

- Interactive maps with navigation and routing (Mapbox integration)
- User authentication and account management
- Location-based forums for reporting congestion, hazards, and city events
- Dashboard displaying city activity and community contributions

Excluded Features (Prototype Limitations):

- Full-scale analytics and predictive AI features
- Real-time alerts for all city events
- Enterprise-level scalability

The system improves commute efficiency, enhances situational awareness, and supports community engagement in urban areas.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
API	Application Programming Interface, a set of rules for software interaction
MFCD	Major Functional Component Diagram
RWP	Real World Product
IoT	Internet of Things, systems combining sensing, computing, communication
Crowdsourced Reporting	Gathering information via user contributions

Term	Definition
Privacy-by-Design	Ensuring personal data is minimized and protected by default
Mapbox	Third-party mapping API for navigation and routing
ElysiaJS	Backend framework for handling API requests and business logic

1.4. References

[1] J. Connelly, “Lab 1 – Smart City Product Description,” CS411W, Old Dominion University, 2026.

[2] Affinity Studio, “Web & App Development,” 2025. [Online]. Available: https://www.affinity.studio/

[3] GitHub, “Where the world builds software,” 2025. [Online]. Available: https://github.com/

[4] React Native, “React Native Documentation,” 2025. [Online]. Available: https://reactnative.dev/

[5] Firebase, “Firebase Documentation,” 2025. [Online]. Available: https://firebase.google.com/

[6] Mapbox, “Mapbox API Documentation,” 2025. [Online]. Available: https://docs.mapbox.com/

[7] OpenStreetMap Wiki, “API & Overpass API,” 2024. [Online]. Available: https://wiki.openstreetmap.org/

[8] Expo, “Expo Documentation,” 2025. [Online]. Available: https://expo.dev/

1.5. Overview

The remainder of this SRS includes:

- Section 2 – Overall Description: System architecture, functional components, user characteristics, constraints, and dependencies.

- Section 3 (not included yet): Detailed functional requirements for developers.

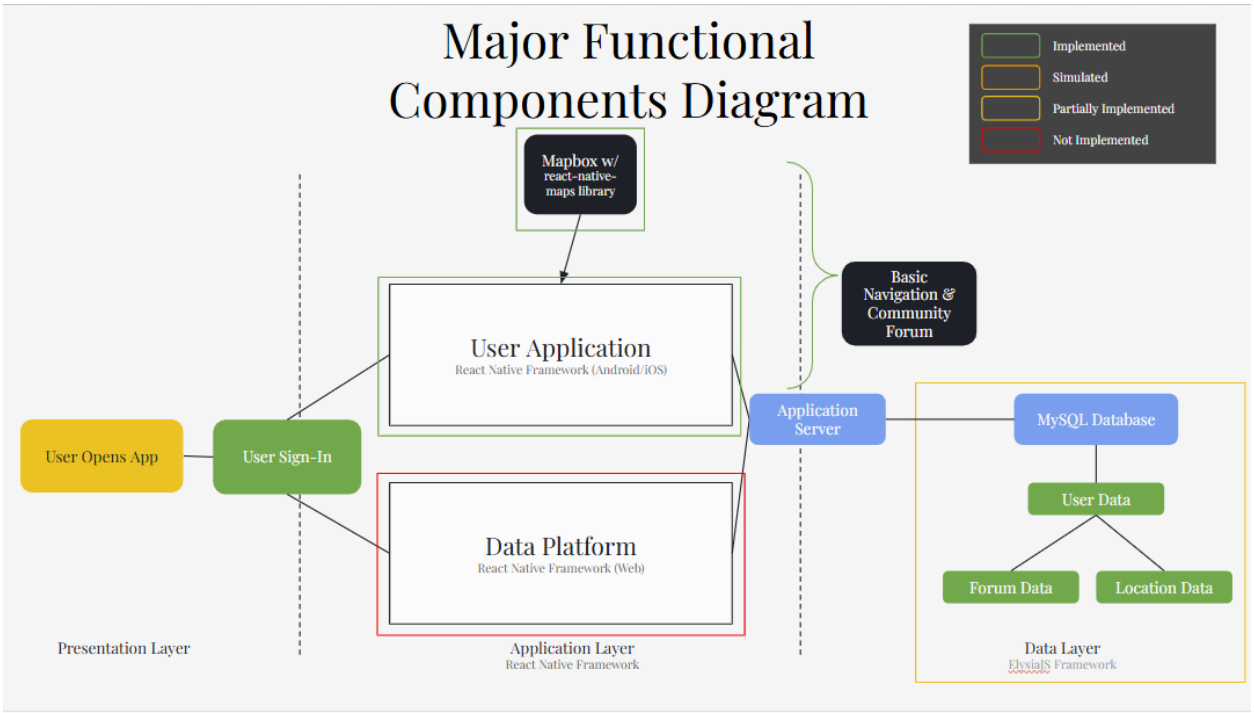
2. Overall Description

2.1. Product Perspective

Smart City integrates navigation services, user authentication, and community-driven information into a single platform. The system interacts with external services like **Mapbox** for mapping and routing, and **Firebase** for user account management.

Architecture Layers:

- **Presentation Layer:** React Native (Mobile), Web interface
- **Application Layer:** ElysiaJS backend handles business logic
- **Data Layer:** MySQL database for user, forum, and location data



Component Interactions:

- Users interact with mobile/web frontend

- Frontend communicates with backend for authentication, forum posts, and map updates
- Backend queries MySQL and external APIs
- Forum and navigation data are aggregated in the data layer for dashboard visualization

2.2. Product Functions

Feature	Prototype Implementation
Login/Authentication	Fully Implemented
Account Creation/Deletion	Partially Implemented
Access Community Forum	Partially Implemented
Map Navigation / Dynamic Routing	Partially Implemented
Alerts and Notifications	Partially Implemented
Parking Location Updates	Not Implemented
Dashboard Analytics	Partially Implemented
Data Anonymity	Fully Implemented

2.3. User Characteristics

User Role	Skill Level	Usage Scenario
General User	Basic smartphone experience	View maps, receive routing info, report hazards
Moderator	Intermediate	Manage forum posts, moderate content
Admin	Advanced	Maintain system, analyze aggregated data

2.4. Constraints

- Must operate on **iOS and Android** devices
- Uses **React Native**, **ElysiaJS**, and **MySQL**
- Depends on **Mapbox API** for navigation

- Prototype is limited to semester development time

2.5. Assumptions and Dependencies

- Internet connectivity is required
- Firebase accounts for authentication
- Mapbox and OpenStreetMap APIs available
- MySQL database hosted on university VM